

p-Block Elements Complete High-Yield Notes

Group 15, 16, 17 and 18 Comprehensive Study Sheet

Chapter / Focus Area: Inorganic Chemistry Class 12

1. Group 15 (Nitrogen Family)

- Anomalous behavior of Nitrogen due to small size, high electronegativity, and absence of d-orbitals.
 - Oxides of Nitrogen: N_2O (laughing gas, neutral), NO (neutral), NO_2 (brown gas, acidic).
 - Haber's Process for Ammonia: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (Fe catalyst, Mo promoter, 700 K, 200 atm).
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2. Group 16 & 17 Trends

- Contact Process for H_2SO_4 : V_2O_5 catalyst at 720 K and 2 bar pressure.
 - Anomalous behavior of Fluorine: Lower electron gain enthalpy than Chlorine due to interelectronic repulsions in small 2p orbital.
 - Oxoacids of Halogens & Bleaching action of Cl_2 via oxidation.
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Official Verified Study Resource • Mentored by Ajay Choudhary Sir

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